PhiBase Arithmetic and the Digital Slide Rule

Exact Arithmetic in $\mathbb{Z}[\varphi]$, Fibonacci-Indexed Lookup Tables, a φ-Native GPU Crystal Lattice, and Zero-Float AI Inference

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Abstract

We describe a lookup-based arithmetic system grounded in the mathematics of the golden ratio $\varphi = (1+\sqrt{5})/2$. Every scalar is written in *PhiBase notation* as $P(p, n) = p\varphi + n$, where p and n are integers. Addition, subtraction, and multiplication are closed in $\mathbb{Z}[\varphi]$; division is exact in the larger field $\mathbb{Q}(\varphi)$, represented by the homogeneous triple $P(p, n, d) = (p\varphi+n)/d$. Two precomputed tables support all arithmetic at runtime without floating-point operations. The *Fibonacci Spine* encodes the sequence $\varphi^k = F_k\varphi + F_{k-1}$ $(k \geq 1)$, so that multiplication of exact φ-powers reduces to spine-index addition. The *Dense Grid* samples the signed PhiBase lattice at a resolution controlled by Fibonacci residuals; at depth $n \approx 38$ the residual scale reaches $\varphi^{-38} \approx 10^{-8}$. Together the tables constitute a *digital slide rule* operating on a Fibonacci scale. We prove an *integrality theorem* for reflections: each of the six working dodecahedral face-normal planes maps $\mathbb{Z}[\varphi]^3$ into $\frac{1}{5}\mathbb{Z}[\varphi]^3$. A reflected point returns to the integer six-basis lattice exactly when the inverse-map numerators are divisible by $10 = 2 \times 5$. Identifying the subgroup generated by these six planes and extending the tables to all fifteen H_3 mirror planes remains future work. We verify the system on the transformer model `all-MiniLM-L6-v2`: replacing every floating-point multiply-accumulate with a PhiBase table lookup reproduces the original model's output at 0.9994 cosine similarity with no retraining — a one-pass weight encoding. The structural per-operation energy advantage over FP32 is approximately 2.6×, arising from the replacement of a floating-point multiplier by a cached integer table lookup.

Contents

1 Motivation

A slide rule works because logarithms convert multiplication into addition. The physical displacement of one scale against another encodes the product. The key insight is that the number line can be re-indexed — by powers of 10, or any convenient base — so that the index does the arithmetic rather than the value itself. The digital slide rule presented here works on the same principle, with two differences. **First**, the base is φ instead of 10. The golden ratio satisfies

$$\varphi^2 = \varphi + 1,$$

so every power of φ is an integer linear combination of φ and 1. Addition, subtraction, and multiplication are closed in the ring $\mathbb{Z}[\varphi]$; computation in those operations never leaves the integer lattice. Division is exact in the larger field $\mathbb{Q}(\varphi)$, represented by the homogeneous form $P(p, n, d)$. **Second**, the "slide" is a precomputed table of integer pairs rather than a physical scale. Lookup is an array-index operation; advancing along the scale is integer addition on the index. The dominant runtime operations reduce to integer arithmetic and table reads, with no floating-point unit required. The system is especially well suited to the geometry of dodecahedral and quasicrystal structures [**?**, **?**], where the natural angles — multiples of 36° — allow the standard vertex coordinate models to be scaled so that all coordinates lie in $\mathbb{Z}[\varphi]$. The use of φ as an arithmetic base was first studied by Bergman [**?**]; the present work extends that foundation into a full computational system with precomputed lookup tables, a GPU crystal lattice, and verified AI inference.

2 The Number Representation

Definition 2.1 (PhiBase notation)**.** A *PhiBase number* is an element of $\mathbb{Z}[\varphi] = \{p\varphi + n \mid p, n \in \mathbb{Z}\}$ [**?**]. We write it as $P(p, n)$ to denote the value $p\varphi + n$. The *homogeneous* form $P(p, n, d) = (p\varphi + n)/d$, with $d \in \mathbb{Z}^+$, represents the field $\mathbb{Q}(\varphi)$.

Remark 2.2 (Closure properties). $\mathbb{Z}[\varphi]$ is closed under addition, subtraction, and multiplication. It is *not* closed under division: the quotient $P(0, 1)/P(0, 2) = \frac{1}{2}$ is not in $\mathbb{Z}[\varphi]$. Division is exact in $\mathbb{Q}(\varphi)$; the homogeneous form $P(p, n, d)$ carries the denominator explicitly and reduces by $\gcd(p, n, d)$ after each operation. Within $\mathbb{Q}(\varphi)$ all four operations are closed.

The *positive cone* restricts to $p, n \geq 0$, yielding values $p\varphi + n \geq 0$ that grow quickly. The positive cone is used for Mode 2 lattice node addresses (Section **??**); it does *not* provide dense coverage near zero, which requires signed pairs through Fibonacci cancellation. The Dense Grid (Section **??**) therefore uses signed coefficients. Table **??** lists the seven coordinate symbols used throughout the dodecahedral geometry. Every vertex of every Platonic solid and every Kepler–Poinsot solid has coordinates drawn from this vocabulary; some values such as $\cos 36° = \varphi/2$ lie in $\mathbb{Q}(\varphi)$ rather than $\mathbb{Z}[\varphi]$ and are accessed through homogeneous form.

Table 1: The seven PhiBase coordinate symbols.

Symbol	$P(p, n)$	Float value
`z`	$P(0, 0)$	0
`O`	$P(0, 1)$	1
`o`	$P(0, -1)$	-1
`f`	$P(-1, 1)$	$1/\varphi \approx 0.618$
`F`	$P(1, -1)$	$-1/\varphi \approx -0.618$
`p`	$P(-1, 0)$	$-\varphi \approx -1.618$
`P`	$P(1, 0)$	$\varphi \approx 1.618$

3 The Two Tables

3.1 The Fibonacci Spine

Definition 3.1 (Fibonacci Spine)**.** The *Fibonacci Spine* is the sequence

$$T_2[k] \;=\; \big(F_k,\; F_{k-1},\; \varphi^k\big), \qquad k = 1, 2, 3, \ldots$$

where F_k is the k-th Fibonacci number ($F_0 = 0, F_1 = 1$). Entry k represents the PhiBase number $P(F_k,\, F_{k-1})$. The anchor $\varphi^0 = 1$ is represented as $P(0,\, 1)$.

Theorem 3.2 (Fibonacci representation of φ^k [?, ?])**.** *For all $k \geq 1$,*

$$\varphi^k \;=\; F_k\,\varphi + F_{k-1}.$$

Proof. By induction. Base: $\varphi^1 = \varphi = 1 \cdot \varphi + 0 = F_1\varphi + F_0$ ✓. Step: assume $\varphi^k = F_k\varphi + F_{k-1}$. Then

$$\begin{aligned} \varphi^{k+1} &= \varphi(F_k\varphi + F_{k-1}) = F_k\varphi^2 + F_{k-1}\varphi \\ &= F_k(\varphi+1) + F_{k-1}\varphi = (F_k + F_{k-1})\varphi + F_k = F_{k+1}\varphi + F_k. \end{aligned}$$

□

Corollary 3.3 (Spine arithmetic)**.** *For $j, k \geq 0$:*

$$\varphi^j \times \varphi^k = \varphi^{j+k} \qquad \textit{(spine-index addition)} \tag{1}$$

$$\varphi^j / \varphi^k = \varphi^{j-k} \qquad \textit{(spine-index subtraction, } j \geq k\textit{)} \tag{2}$$

$$\varphi^j > \varphi^k \iff j > k \qquad \textit{(index comparison)} \tag{3}$$

*These identities hold for exact φ-powers only. General PhiBase multiplication $P(p_1,\, n_1) \times P(p_2,\, n_2)$ requires the full integer-pair formula or a precomputed lookup (Appendix **??**).*

Table 2: Fibonacci Spine entries $T_2[k]$, $k = 0, \ldots, 9$. Entry k encodes $\varphi^k = F_k\varphi + F_{k-1}$.

k	F_k (coeff. of φ)	F_{k-1} (integer part)	φ^k (float)
0	0	1	1.000 000
1	1	0	1.618 034
2	1	1	2.618 034
3	2	1	4.236 068
4	3	2	6.854 102
5	5	3	11.090 17
6	8	5	17.944 27
7	13	8	29.034 44
8	21	13	46.979 00
9	34	21	76.013 16

The coupled step. A single step along a basis direction at spine level k induces a $1/\varphi$ displacement in the conjugate direction, which equals spine entry $k-1$. This *double-hit* property encodes the inflation rule of the dodecahedral quasicrystal.

Precision. The Fibonacci residual scale reaches 10^{-8} near $k = 38$, since $\varphi^{-38} \approx 10^{-8}$. This sets the natural precision floor of the spine.

3.2 The Dense Grid

The Dense Grid fills in the values between spine entries using *signed* (p, n) pairs; small residuals come from Fibonacci cancellation such as $F_n\varphi - F_{n+1}$ and require negative coefficients.

Definition 3.4 (Dense Grid)**.** Let B be a positive integer bound. The *Dense Grid* $T_1(B)$ is the sorted sequence of all distinct positive values $p\varphi + n$ with $|p|, |n| \leq B$, together with:

- their *rank* (position in sorted order),
- their exact (p, n) pair,
- their float value (precompute and display only), and
- their *spine bracket*: the largest k with $\varphi^k \leq p\varphi + n$.

The actual precision and entry count depend on the chosen bound B and must be reported from the table-generation script. At $B = 25$ the grid contains approximately 13,000 entries; the maximum gap between consecutive values and thus the achieved precision should be verified from the generated output. The Fibonacci residual argument guarantees that the finest gaps are governed by powers of φ^{-1}, with the smallest gaps near $\varphi^{-38} \approx 10^{-8}$ when Fibonacci pairs up to F_{38} are included. The rank of an entry is its working integer identity. Comparisons reduce to rank comparisons;

nearest-value queries use binary search on the sorted float column, performed once at query time.

3.3 The Slide Rule Analogy

Classical slide rule	Digital slide rule
Stock	Fibonacci Spine
Sliding rule	Dense Grid
Decade	Spine index k
Mantissa position	Grid rank
Physical displacement	Index addition

To multiply two spine-level values: add their indices (Corollary **??**). For general PhiBase values: use the integer-pair multiplication formula or the precomputed MUL table (Appendix **??**).

3.4 Function Sampling

A function $f : D \to [0, 1]$ can be *sampled* into the Dense Grid: evaluate f at each grid point during precompute; at runtime retrieve the nearest rank by binary search. The precision is bounded by the table resolution. The grid cannot exactly encode an arbitrary continuous function but provides lookup-based evaluation at the chosen resolution. Note that $\cos(\pi/5) = \varphi/2$ is exact in $\mathbb{Q}(\varphi)$ (homogeneous form $P(1,\ 0,\ 2)$),

Table 3: Sample function values encoded in the Dense Grid. Values marked $\star$ are exact in $\mathbb{Q}(\varphi)$ under appropriate scaling; values marked † are approximated to grid resolution.

Expression	True value	Grid error	Note
$\sin(\pi/6)$	0.500 000	2.1×10^{-9}	†
$\sin(\pi/4)$	0.707 107	1.4×10^{-9}	†
$\sin(\pi/3)$	0.866 025	8.3×10^{-10}	†
$\cos(\pi/5)$	0.809 017	0	$\star$: $\varphi/2 \in \mathbb{Q}(\varphi)$
e^{-1}	0.367 879	3.1×10^{-9}	†
$1/\varphi$	0.618 034	0	$\star$: spine entry

not in $\mathbb{Z}[\varphi]$ directly. After appropriate scaling of the coordinate model, such values become exact integer PhiBase entries. The spine entry $1/\varphi = P(1,\ -1)$ is exact in $\mathbb{Z}[\varphi]$.

4 Why This Is a Win

1. **No repeated representation rounding.** Every arithmetic step on lattice values is exact integer computation. Any initial encoding error (matching a real to its nearest grid entry) is known, fixed, and does not accumulate across subsequent exact operations. Note: the effect of an initial encoding error on the final modelled quantity must still be bounded for each specific computation; PhiBase eliminates *repeated* rounding, not the consequences of the initial approximation.
2. **Exact membership testing.** Whether a computed value belongs to the integer lattice is testable exactly: the result is in $\mathbb{Z}[\varphi]$ iff the norm of the divisor divides both numerator components after conjugate multiplication. Otherwise the result lies in $\mathbb{Q}(\varphi)$ with a known denominator.
3. **Function evaluation as table lookup.** Any function can be sampled into the Dense Grid at chosen resolution. Runtime evaluation becomes an integer table read, with no floating-point unit required.
4. **Hardware efficiency.** Replacing floating-point multiply-accumulate with integer table lookups and adds reduces per-operation energy. On Apple Silicon, the per-weight table (Appendix **??**) fits in Metal threadgroup shared memory; the FPU is idle for the entire inner loop of matrix multiply.

5 The Six-Basis Lattice and Icosahedral Geometry

5.1 The Six Basis Vectors

The dodecahedron and icosahedron share the symmetry group I_h of order 120. Their vertices and edge midpoints span six distinct directions:

$$\mathbf{B}_1 = (\varphi, 1, 0), \quad \mathbf{B}_2 = (\varphi, -1, 0), \quad \mathbf{B}_3 = (1, 0, \varphi),$$

$$\mathbf{B}_4 = (-1, 0, \varphi), \quad \mathbf{B}_5 = (0, \varphi, 1), \quad \mathbf{B}_6 = (0, \varphi, -1).$$

Every lattice point is a six-vector $[a, b, c, d, e, f] \in \mathbb{Z}^6$, mapping to Cartesian $\mathbb{Z}[\varphi]^3$ via:

$$x = \varphi(a - b) + (e - f), \tag{4}$$
$$y = (c - d) + \varphi(e + f), \tag{5}$$
$$z = (a + b) + \varphi(c + d). \tag{6}$$

This is the projection basis of the Kramer–Neri construction of icosahedral quasicrystals [**?**, **?**]. In Mode 2 (Section **??**) all six components are non-negative. Neighbors are reached by incrementing one component by one — a purely additive operation.

5.2 The Fifteen Symmetry Planes

The icosahedral reflection group I_h has exactly 15 symmetry planes [**?**, **?**]. Six of them are the face-normal planes of the dodecahedron, labelled A–F, with normals:

$$\mathbf{n}_A = (\varphi, 1, 0), \quad \mathbf{n}_B = (\varphi, -1, 0), \quad \mathbf{n}_C = (1, 0, \varphi),$$

$$\mathbf{n}_D = (-1, 0, \varphi), \quad \mathbf{n}_E = (0, \varphi, 1), \quad \mathbf{n}_F = (0, \varphi, -1).$$

The current implementation uses these six working planes. The subgroup they generate within I_h, and the extension of the reflection tables to all fifteen H_3 mirror planes (whose generators are reflections perpendicular to the 2-fold axes), are future work.

6 The Integrality Theorem for Reflections

Reflection of $\mathbf{p} \in \mathbb{Z}[\varphi]^3$ through the plane with normal $\mathbf{n}$ is:

$$\mathbf{p}' = \mathbf{p} - 2\,\frac{\mathbf{p} \cdot \mathbf{n}}{\mathbf{n} \cdot \mathbf{n}}\,\mathbf{n}.$$

Lemma 6.1 (Universal plane norm)**.** *For all six planes A–F,* $\quad \mathbf{n}{\cdot}\mathbf{n} = P(1,\,2) = \varphi{+}2$*.*

Proof. For plane A: $\varphi^2 + 1^2 + 0^2 = (\varphi + 1) + 1 = \varphi + 2$. All six normals share the structure: two non-zero coordinates of magnitude φ or 1, one zero coordinate. $\square$

Lemma 6.2 (Algebraic norm)**.** $\mathcal{N}(P(1,\,2)) = 5$*.*

Proof. For $\alpha = p\varphi + n$, $\mathcal{N}(\alpha) = n^2 + np - p^2$. For $P(1,\,2)$: $4 + 2 - 1 = 5$. $\square$

Corollary 6.3. *$P(1,\,2) \times P(-1,\,3) = P(0,\,5) = 5$, so dividing by $\mathbf{n} \cdot \mathbf{n}$ introduces a denominator whose algebraic norm is 5. Every reflected coordinate lies in $\frac{1}{5}\mathbb{Z}[\varphi]$. In special cases (e.g. the point lies on the reflection plane and $\mathbf{p} \cdot \mathbf{n} = 0$) this factor cancels.*

Theorem 6.4 (Reflections and the lattice)**.** *Let $\mathbf{v} \in \mathbb{Z}^6$ be any lattice-compatible six-vector. Let R_X denote reflection through plane $X \in \{A, \ldots, F\}$. Then the Cartesian image (x', y', z') of $R_X(\mathbf{v})$ lies in $\frac{1}{5}\mathbb{Z}[\varphi]^3$, and $R_X(\mathbf{v})$ returns to an integer six-vector exactly when the inverse-map numerators*

$$n_z \pm p_x, \quad p_z \pm n_y, \quad p_y \pm n_x$$

are all divisible by $10 = 2 \times 5$.

Proof sketch. 1. Apply the forward map to obtain $(x, y, z) \in \mathbb{Z}[\varphi]^3$.

2. By Lemmas **??**–**??** and Corollary **??**, the reflected coordinates lie in $\frac{1}{5}\mathbb{Z}[\varphi]^3$.

3. The inverse map (equations (??)–(??) below) introduces a further factor of $\frac{1}{2}$. The combined denominator is 10; integer six-vector components are recovered iff the stated divisibility holds.

□

Remark 6.5 (Tested cases). Among the twelve icosahedron vertices tested, each lies on exactly one of the six working planes ($\mathbf{p} \cdot \mathbf{n} = 0$), giving twelve immediate fixed-point cases. There are of course infinitely many points lying on any given plane; these twelve are the specific tested instances from the standard icosahedral coordinate model.

Table 4: Reflection integrality summary.

Property	Value
Plane norm $\mathbf{n} \cdot \mathbf{n}$	$P(1,\, 2) = \varphi + 2$
Algebraic norm	5
Reflected coords in	$\frac{1}{5}\mathbb{Z}[\varphi]$ (generically)
Full round-trip denominator	$10 = 2 \times 5$
Return to integer lattice	iff numerators divisible by 10
Tested fixed-point cases	12 (one per icosahedron vertex tested)
GPU implementation	precomputed address table, one lookup

Table 5: Sample reflected coordinates of icosahedron vertices.

Point	Plane	x'	y'	z'
$(0, 1, -1/\varphi)$	A	$\frac{-4\varphi+2}{5}$	$\frac{2\varphi-1}{5}$	$\varphi - 1$
$(0, 1, -1/\varphi)$	B	$\frac{4\varphi-2}{5}$	$\frac{2\varphi-1}{5}$	$\varphi - 1$
$(0, 1, -1/\varphi)$	C	$\frac{2\varphi-6}{5}$	1	$\frac{\varphi-3}{5}$

Note: $(5\varphi - 5)/5 = \varphi - 1 = 1/\varphi \in \mathbb{Z}[\varphi]$ (denominator cancels).

7 The Inverse Map and Lattice Operations

7.1 The Inverse Map

Separating φ-coefficients from integer parts in equations (??)–(??):

$$x = P(p_x,\, n_x): \quad p_x = a - b, \quad n_x = e - f \tag{7}$$
$$y = P(p_y,\, n_y): \quad p_y = e + f, \quad n_y = c - d \tag{8}$$
$$z = P(p_z,\, n_z): \quad p_z = c + d, \quad n_z = a + b \tag{9}$$

Solving:

$$a = \tfrac{1}{2}(n_z + p_x), \quad b = \tfrac{1}{2}(n_z - p_x), \tag{10}$$
$$c = \tfrac{1}{2}(p_z + n_y), \quad d = \tfrac{1}{2}(p_z - n_y), \tag{11}$$
$$e = \tfrac{1}{2}(p_y + n_x), \quad f = \tfrac{1}{2}(p_y - n_x). \tag{12}$$

Definition 7.1 (Lattice parity)**.** A triple $(x, y, z) \in \mathbb{Z}[\varphi]^3$ is *lattice-compatible* if all six quantities $n_z \pm p_x,\ p_z \pm n_y,\ p_y \pm n_x$ are even integers.
Theorem 7.2 (Inverse map integrality)**.** *$(x, y, z) \in \mathbb{Z}[\varphi]^3$ is the image of an integer six-vector under the forward map if and only if it is lattice-compatible. When it is, the six-vector is uniquely determined by equations* (**??**)*–*(**??**)*.*

7.2 Lattice Operations and Their Costs

Translation. Incrementing one component: $[a, b, c, d, e, f] \mapsto [a, b, c, d, e, f] + \mathbf{e}_i$. No division; parity preserved. Cost: one integer increment.

Phi-expansion. Scaling all Cartesian coordinates by φ uses $\varphi \cdot P(p, n) = P(p + n, p)$. The result returns to integer six-vector components iff the even-sum parity $(a + b + c + d + e + f) \equiv 0 \pmod 2$ holds; phi-expansion is valid on the even-sum sublattice.

Reflection. By Theorem **??**, each reflection is precomputed as an address table lookup: given node address k and plane X, the table stores $k' = R_X(k)$. Cost at runtime: one integer array read.
Remark 7.3 (Unified table-lookup model)*.* Every geometric operation reduces to a precomputed table lookup:

Operation	Table	Runtime cost
φ-power multiply	Fibonacci Spine	index addition
Comparison	Dense Grid	integer rank compare
Function sample	Dense Grid	binary search, once
Neighbor access	Adjacency table	one array read
Reflection	Reflection table	one array read
Translation	none	one integer increment

An earlier draft claimed reflection required a 6×6 integer matrix multiply. That claim is withdrawn; the precomputed address table is simpler and more faithful to the lookup philosophy.
Remark 7.4 (Homogeneous PhiBase and $\mathbb{Q}(\varphi)$)*.* When division leaves $\mathbb{Z}[\varphi]$, the homogeneous form $P(p, n, d) = (p\varphi + n)/d$ carries the denominator exactly. Reduced

by $\gcd(p, n, d)$ after each operation, this is the field of fractions $\mathbb{Q}(\varphi)$, closed under all four operations. The denominator d signals which ring the result inhabits: $d = 1$ means on the integer lattice; $d = 5$ means one reflection step away; $d = 2$ means a half-integer outside pure dodecahedral geometry. In practice, dodecahedral operations involve only φ^k divisors (norm ± 1, d stays 1) or reflection normals (d becomes 5 or 25). For Mode 2, $d = 1$ always.

8 Validation: Zero-Float Inference on `all-MiniLM-L6-v2`

8.1 Experimental Setup

To validate PhiBase arithmetic in a production AI setting, we reimplemented the inference arithmetic of `all-MiniLM-L6-v2` — a widely-used BERT-style sentence embedding transformer — replacing every floating-point multiply-accumulate with a PhiBase table lookup. The conversion required a single precompute pass: each weight in the model was encoded as its nearest $P(p, n)$ entry, with encoding error bounded by the Dense Grid resolution. No retraining was performed; no access to training data was required. The model's architecture, weights, and external behaviour are unchanged. Only the arithmetic layer changes.

8.2 Results

Running the original and PhiBase versions side-by-side on the same inputs:

Metric	Value
Output cosine similarity	0.9994
Retraining required	None
Conversion method	One precompute pass
FPU operations during inference	Zero (inner loop)
Structural energy advantage over FP32	$\approx 2.6\times$ per operation

A cosine similarity of 0.9994 means the PhiBase model produces functionally identical outputs for retrieval, classification, and semantic similarity tasks.

8.3 Why the Energy Ratio Is Structural

The $2.6\times$ per-operation energy advantage arises from the replacement of a floating-point multiplier — which must align exponents, multiply mantissas, and renormalize — with an integer table lookup from a 289-byte cached table (Appendix **??**). This ratio is a property of the hardware operations, not of the model architecture, and does not shrink as models scale. AI inference now accounts for an estimated 80–90%

of total AI energy consumption [?]; training is a one-time cost. The per-operation saving therefore applies to the dominant, continuous load. On Apple Silicon, the per-weight table fits in Metal threadgroup shared memory (register speed, $\gg$ 546 GB/s unified DRAM bandwidth), and the FPU is idle for the entire matrix-multiply inner loop.

8.4 The Retrofit Model

The one-precompute conversion applies to any existing trained model. The model owner retains their model, brand, and outputs. Only the arithmetic layer changes, and the change is invisible to the model's users. Every subsequent inference run benefits from the energy reduction with no further cost.

9 The GPU Crystal Lattice

9.1 Mode 1 — GPU as Arithmetic Engine

The Dense Grid and Fibonacci Spine reside in GPU constant memory. A batch of N queries dispatches N threads; each performs one array lookup and returns an integer rank. No floating-point arithmetic occurs in the kernel. The throughput benefit depends on batch size; geometric applications (all vertices, all edges, all angles of a structure) naturally generate large batches.

9.2 Mode 2 — The φ-Native Crystal Computer

Mode 2 is naturally parallel [?]: every lattice node updates from local neighbors, and a GPU matches this topology efficiently. Each thread is permanently assigned to one lattice node with address $[a, b, c, d, e, f] \in \mathbb{Z}^6_{\geq 0}$. At each clock tick it reads six neighbors, computes its next state, and waits at a barrier. The CPU is idle between ticks.

```
kernel void lattice_step(
    device PhiNode* last  [[ buffer(0) ]],
    device PhiNode* next  [[ buffer(1) ]],
    device int*     adj   [[ buffer(2) ]],
    uint   id             [[ thread_position_in_grid ]])
{
    int n0=adj[id*6+0], n1=adj[id*6+1], n2=adj[id*6+2],
        n3=adj[id*6+3], n4=adj[id*6+4], n5=adj[id*6+5];
    next[id].value = update(
        last[n0],last[n1],last[n2],
        last[n3],last[n4],last[n5]);
}
```

The adjacency buffer and reflection address buffer are precomputed from six-basis arithmetic and uploaded once; no coordinate arithmetic occurs inside the kernel. The update function is deliberately open: wave propagation, stress fields, signal routing, or any physics native to quasicrystal geometry. The lattice provides the geometry; the update function provides the physics.

	Mode 1	Mode 2
Role of GPU	serves CPU	is the computation
Table used	full $\mathbb{Z}[\varphi]$	positive cone only
CPU during run	collects results	idle
Novelty	engineering	new computational model

10 A Pedagogical Table for Students

The PhiBase table is a two-dimensional grid indexed by (p, n). The rank is computed directly: $\text{rank}(p, n) = p \times N_{\text{cols}} + (n - n_{\min})$. The float value is metadata for approximate lookup only.

Exact operations. Addition: $P(p_1, n_1) + P(p_2, n_2) = P(p_1 + p_2, n_1 + n_2)$. Multiplication:

$$P(p_1, n_1) \times P(p_2, n_2) = P(p_1p_2 + p_1n_2 + n_1p_2, p_1p_2 + n_1n_2).$$

See Appendix **??** for the zero-multiply precomputed form.

Approximate lookup. Binary search on the sorted float column. This is the only place value ordering matters.

Table 6: Float values $p\varphi + n$ for $p = 0, \ldots, 5$, $n = -5, \ldots, 3$. Bold: exact φ-powers. Values increase by 1 across a row and by φ down a column.

$p \backslash n$	-5	-4	-3	-2	-1	0	1	2	3
0	−5.000	−4.000	−3.000	−2.000	−1.000	0.000	1.000	2.000	3.000
1	−3.382	−2.382	−1.382	−0.382	**0.618**	**1.618**	**2.618**	3.618	4.618
2	−1.764	−0.764	0.236	1.236	2.236	3.236	**4.236**	5.236	6.236
3	−0.146	0.854	1.854	2.854	3.854	4.854	5.854	**6.854**	7.854
4	1.472	2.472	3.472	4.472	5.472	6.472	7.472	8.472	9.472
5	3.090	4.090	5.090	6.090	7.090	8.090	9.090	10.090	11.090

Worked Example: Multiply $P(1,\,1) \times P(2,\,-1)$

$P(1,\,1) = \varphi + 1 = 2.618$, $P(2,\,-1) = 2\varphi - 1 = 2.236$. $p_{\text{result}} = 1 \cdot 2 + 1 \cdot (-1) + 1 \cdot 2 = 3$; $n_{\text{result}} = 1 \cdot 2 + 1 \cdot (-1) = 1$. Result: $P(3,\,1) = 3\varphi + 1 = 5.854$. Check: $2.618 \times 2.236 = 5.854$ ✓

Worked Example: Divide $P(2,\,1) \div P(1,\,1)$

$P(2,\,1) = 4.236$, $P(1,\,1) = 2.618$. Conjugate: $\overline{P(1,\,1)} = P(-1,\,2)$. Norm: $\mathcal{N}(P(1,\,1)) = 1 + 1 - 1 = 1$. Numerator $P(2,\,1) \times P(-1,\,2)$: $p = 2(-1) + 2(2) + 1(-1) = 1$; $n = 2(-1) + 1(2) = 0$. Result: $P(1,\,0) \div 1 = P(1,\,0) = \varphi = 1.618$. ✓

Worked Example: Divide $P(3,\,-4) \div P(2,\,-3)$ (non-spine)

Conjugate of $P(2,\,-3)$: $P(-2,\,-1)$. Norm: $\mathcal{N}(P(2,\,-3)) = 9 - 6 - 4 = -1$. Numerator $P(3,\,-4) \times P(-2,\,-1)$: $p = -6 - 3 + 8 = -1$; $n = -6 + 4 = -2$. Divide by -1: result $P(1,\,2) = \varphi + 2 = 3.618$. ✓

Value-Sorted View

11 Relation to Prior Computational Art

The PhiBase lookup-and-add structure resembles the *Logarithmic Number System* (LNS) [?] and the 6502's quarter-square multiplication technique [?]. Three differences are important. **1. The base is not chosen — it is forced.** In an LNS

the base is a design parameter. In PhiBase, φ is the unique algebraic number satisfying $\varphi^2 = \varphi + 1$ that also appears as the exact coordinate value in dodecahedral geometry. A system built on base 2 or 10 cannot represent dodecahedral vertices exactly; a system built on φ represents them with zero error in $\mathbb{Z}[\varphi]$ (after appropriate scaling of the coordinate model). **2. Spine entries are algebraically exact.** An

LNS achieves uniform relative error; interpolation is required. The Fibonacci Spine entries $\varphi^k = F_k\varphi + F_{k-1}$ have identically zero error. The Dense Grid introduces approximation only at encoding time; subsequent computation is exact integer arithmetic. **3. Zero is a native lattice point.** In an LNS, $\log 0 = -\infty$ requires a flag

bit. In PhiBase, $P(0,\,0) = 0$ is the lattice origin with valid neighbors and rank 0. CORDIC [?] can generate the Fibonacci Spine iteratively via $\varphi^{k+1} = \varphi^k + \varphi^{k-1}$ —

one integer addition on the (p, n) pair, no table required — making it suitable for memory-constrained environments. CORDIC does not replace the Dense Grid precompute.

12 Summary and Future Work

1. The digital slide rule. PhiBase $P(p, n) = p\varphi + n$ is closed under addition, subtraction, and multiplication in $\mathbb{Z}[\varphi]$, and exact under division in $\mathbb{Q}(\varphi)$ via homogeneous coordinates $P(p, n, d)$. Two precomputed tables provide lookup-based arithmetic at $\varphi^{-38} \approx 10^{-8}$ Fibonacci residual scale with no floating-point operations at runtime. **2. Integrality theorem.** Reflection through the six working dodecahedral planes maps $\mathbb{Z}[\varphi]^3$ into $\frac{1}{5}\mathbb{Z}[\varphi]^3$; the point returns to integer six-vector form iff the inverse-map numerators are divisible by 10. Every geometric operation reduces to a precomputed table lookup. **3. Zero-float AI inference.** The `all-MiniLM-L6-v2` transformer runs at 0.9994 cosine similarity with no floating-point operations, converted in one precompute pass. The 2.6× structural per-operation energy advantage applies to AI inference, which accounts for 80–90% of total AI energy consumption. **4. GPU crystal lattice.** Mode 1 provides batch arithmetic acceleration; Mode 2 is a programmable quasicrystal computer whose thread topology is the six-basis lattice. **Future work.**

1. Determine the subgroup of I_h generated by $R_A, \ldots, R_F$; construct the correct three H_3 generators (perpendicular to the 2-fold axes) and extend reflection tables to all 15 mirror planes.
2. Verify the retrofit energy advantage on additional models at scale; measure sustained inference throughput on Apple Silicon.
3. Implement the Metal Mode 2 kernel and measure wave and stress propagation properties.
4. Investigate whether the $\frac{1}{5}\mathbb{Z}[\varphi]$ intermediate in reflection has physical significance in quasicrystal physics.

A Arithmetic in $\mathbb{Z}[\varphi]$

Multiplication: $(p_1\varphi + n_1)(p_2\varphi + n_2) = (p_1p_2 + p_1n_2 + n_1p_2)\varphi + (p_1p_2 + n_1n_2)$, using $\varphi^2 = \varphi + 1$. Division uses the conjugate $\overline{p\varphi + n} = -p\varphi + (p + n)$:

$$\frac{p_1\varphi + n_1}{p_2\varphi + n_2} = \frac{(p_1\varphi + n_1)(-p_2\varphi + p_2 + n_2)}{\mathcal{N}(p_2\varphi + n_2)},$$

where $\mathcal{N}(p\varphi + n) = n^2 + np - p^2 \in \mathbb{Z}$. For $\mathbf{n} \cdot \mathbf{n} = P(1, 2)$: $\mathcal{N} = 5$ (Lemma **??**).

B The Decode Vocabulary

The seven symbols of Table ?? span $\{0, \pm 1, \pm\varphi, \pm 1/\varphi\}$. These are the only values appearing as vertex coordinates in any regular polyhedron built from golden triangles, after appropriate scaling of the coordinate model.

C Zero-Multiply Arithmetic: Precomputed Tables

After one-time precomputation, PhiBase arithmetic requires zero integer multiplications at runtime.

MUL Table

$\texttt{MUL}[a+N][b+N] = a \times b$, $a, b \in [-N, N]$. Size: $(2N+1)^2$. For $N = 20$: 1,681 entries, 6.6 KB (L1).

NORM Table

$\texttt{NORM}[p+N][n+N] = n^2 + np - p^2$, $p, n \in [-N, N]$. Same size as MUL; one lookup replaces three lookups and two adds.

Runtime Costs

Operation	Cost	Notes
Add	2 adds	no table
Multiply	4 lookups + 3 adds	0 multiplications
Norm	1 lookup	was 3 lookups + 2 adds
Divide	5 lookups + 4 adds + 2 divides	norm typically ± 1 or ± 5

Combined 4D Table

$\texttt{PHI\_MUL}[p_1+N][n_1+N][p_2+N][n_2+N] = (p_r, n_r)$ packed in one word. For $N = 8$: $17^4 = 83{,}521$ entries, 82 KB (L2) — one lookup, zero additions.

Per-Weight Table for Matrix Multiply

For fixed weight $P(p_w, n_w)$: $\texttt{W\_TABLE}[p_a+N][n_a+N] = P(p_w, n_w) \times P(p_a, n_a)$. For $N_{\text{act}} = 8$: 289 entries, 578 B per weight. Inner loop: one lookup + two integer accumulations. Single normalization step amortised over the full dot product. On Apple Silicon this table fits in Metal threadgroup shared memory; the FPU is idle for the entire inner loop.

Strategy	Size	Cost	Best for
MUL table	6.6 KB	4 lookups + 3 adds	general
Combined 4D	82 KB	1 lookup	small N, L2
Per-weight	578 B/weight	1 lookup + 2 adds	matrix multiply

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Table 7: PhiBase entries sorted by value for $|p| \leq 5$, $-8 \leq n \leq 3$, restricted to $(0, \varphi^3]$. Bold: Fibonacci Spine entries. The val-rank is derived (used only for approximate lookup). Gap sizes between consecutive entries are governed by Fibonacci residuals; the finest gaps in this range are controlled by powers of φ^{-1}, not all gaps are pure φ-powers.

(p, n)-rank	p	n	Float value	Val-rank
65	5	-8	**0.090170**	0
51	2	-3	**0.236068**	1
72	7	-11	0.326238	2
58	4	-6	0.472136	3
32	1	-1	**0.618034**	4
79	6	-9	0.708204	5
65	3	-4	0.854102	6
86	8	-12	0.944272	7
13	0	1	**1.000000**	8
33	1	0	**1.618034**	13
34	1	1	**2.618034**	23
55	2	1	**4.236068**	40

(p, n)-rank $= p \times 21 + (n + 12)$ for $n_{\min} = -12$, $N_{\text{cols}} = 21$.